Graduation Project 2025/2026

A. Project Information

Project Title	**Embedded PBX System With AI Integration**
Project Track	**Embedded Systems / AI / IoT / Communication Networks**
Department	**Electronics and Communication Department**
Supervisor Name	**Ihab A. Ali**
Industrial partner (if any)	----------

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* Please note that the first name **will be referred** to as the main **contact person** for the whole group.

REVIEW THE FOLLOWING POINTS BEFORE SUBMITTING THE PROPOSALS:

- **Proposals will be evaluated based on:**

  (1) Clarity and validity of objectives,

  (2) Its impact on society/industry,

  (3) Writing and presentation quality,

  (4) Originality,

  (5) Completing ALL the required fields.

- **Review the language of the proposal.**

  It is important that the document is correct from grammatical and vocabulary points of view. Low quality language is one of the major reasons for rejecting proposals.

- **Avoid plagiarism.**

  Never "copy and paste" from a website or a paper. Read from multiple sources and write your own proposal with your original language.

- **References citation.**

  If you used ideas or a sentence from any resource link a website, a paper, or a book, this must be included in the references section. Proposals that don't include the correct references are rejected.

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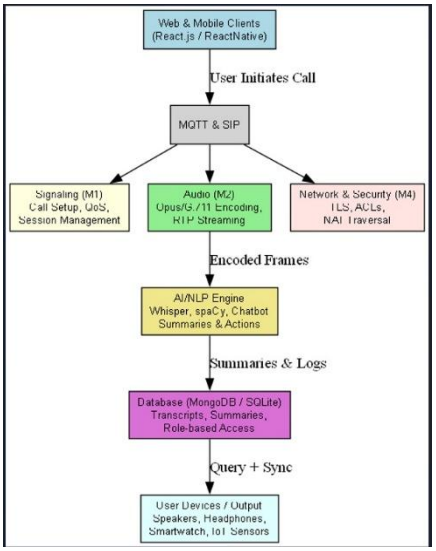
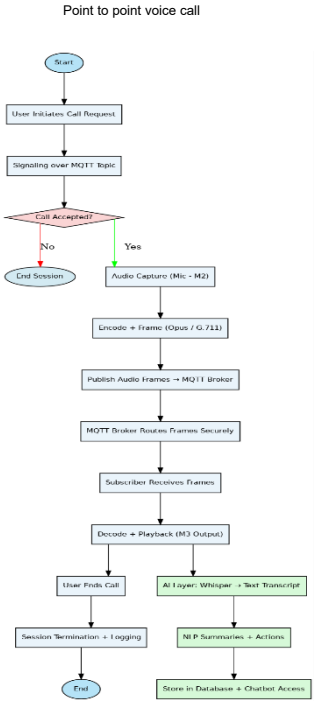
A. PROJECT DESCRIPTION

1. OVERVIEW

(i) Problem definition, (ii) Approach and tools/techniques, and (iii) Overview of system modules.

Block diagram describes the ideas:





(i) Problem Definition:

Modern enterprises and call centers face significant challenges in communication infrastructure. Traditional **PBX and VoIP systems** are limited to voice and video calls, lacking built-in intelligence, role-aware data access, and seamless IoT integration. On the other hand, AI-powered platforms exist but are often **fragmented, expensive, or insecure**, requiring multiple disconnected tools for telephony, transcription, and collaboration.

The problems can be summarized as follows:

1. **Fragmented Communication Tools**
   Organizations rely on separate systems for calling, conferencing, transcription, and note-taking, leading to inefficiency, high costs, and difficulty in managing multiple platforms.

2. **Lack of Intelligent Insights**
   Current PBX/VoIP systems cannot automatically generate meeting summaries, extract action points, or provide role-based access to insights. This results in wasted time on manual documentation and limited productivity.

3. **Security & Privacy Gaps**
   Many existing solutions do not enforce **end-to-end encryption, role-based access control, or adversarial robustness** for AI modules. This exposes enterprises to risks such as call interception, data leakage, and prompt-injection attacks on AI.

4. **Limited IoT Integration**
   Traditional SIP-based telephony does not natively support lightweight IoT devices (smartwatches, mobile apps, sensors), while IoT-friendly protocols like MQTT lack the enterprise-grade reliability and security required for corporate communication.

5. **Scalability & Adaptability Issues**
   Enterprises need systems that can grow with them. Current solutions often lock organizations into rigid architectures that cannot easily integrate new devices, AI models, or security layers

There is a critical need for a secure, intelligent, and unified communication system that combines the strengths of traditional VoIP/PBX (SIP) with IoT-friendly signaling (MQTT), while embedding AI-driven transcription, summarization, and role-based retrieval. Such a system must ensure enterprise-grade security, compliance, and scalability to address the evolving demands of modern organizations.

(ii) Approach and tools/techniques:

Our approach is based on designing a modular, hybrid communication system that integrates VoIP (SIP), IoT messaging (MQTT), Artificial Intelligence (AI), and Security into a unified framework. The system is structured in phases, where each module builds on top of the other to ensure both technical feasibility and scalability.

1. **Hybrid Signaling Model**
   - SIP (Session Initiation Protocol): Used for handling traditional call setup, routing, registration, and interoperability with VoIP/PSTN networks.
   - MQTT (Message Queue Telemetry Transport): Used for lightweight IoT signaling, enabling smart devices such as mobile phones, laptops, and wearables to seamlessly connect.
   - Hybrid Gateway: Bridges SIP and MQTT by mapping SIP methods (e.g., INVITE, BYE) to MQTT JSON structures. This ensures smooth interoperability between enterprise telephony and IoT ecosystems.

   Tools: Asterisk/Kamailio/FreeSWITCH (SIP servers), EMQX or Mosquitto (MQTT broker).

2. **Media Layer & Real-Time Communication**
   - Voice/Video Transmission: Handled using RTP/SRTP and WebRTC for secure conferencing and multimedia support.
   - Encryption: SRTP/DTLS ensures confidentiality of media streams.
   - QoS (Quality of Service): Managed to reduce latency and packet loss during calls.

   Tools: WebRTC, RTP/SRTP libraries, Wireshark for testing and QoS analysis.

3. **AI Integration**
   - Speech-to-Text (STT): Real-time transcription of voice streams using Whisper or Vosk.
   - Natural Language Processing (NLP): Meeting notes summarization, key task extraction, and sentiment analysis using spaCy or Hugging Face Transformers.
   - Chatbot: Role-aware query system that allows employees, managers, or admins to retrieve insights securely.

   Tools: Python (PyTorch/TensorFlow), Whisper/Vosk (STT), spaCy/Transformers (NLP), RASA/Dialogflow (chatbot).

4. **Security Architecture**
   - Transport Security: TLS 1.3 for SIP/MQTT, SRTP/DTLS for media.
   - Authentication: Digest Authentication (SIP), mutual TLS (MQTT), RBAC (role-based access control).
   - Data Protection: AES-256 encryption at rest for transcripts, secure containers for AI pipelines.
   - Monitoring & Intrusion Detection: Fail2Ban, Suricata for anomaly detection, plus VPN segmentation for isolation.

   Tools: OpenSSL, Docker/Kubernetes, Prometheus + Grafana (monitoring), ELK Stack (logging).

5. **Management & Monitoring**
   - Dashboards: Real-time performance, call states, AI insights, and security logs.
   - Testing Environment: Simulations of SIP signaling flows and MQTT topic hierarchies.
   - Scalability Testing: Stress-testing under load using SIPp and MQTT stress tools.

   Tools: Prometheus + Grafana, SIPp, MQTT-Stresser, ELK Stack.

(iii) Overview of system modules:

Signaling & Call Setup

- Manages session initiation and termination using **SIP and MQTT protocols**.
- Defines topic hierarchies and message flow for reliable call signaling.
- Ensures that signaling properly triggers audio sessions in coordination with the audio module.
- Adds reliability features for unstable networks and security using TLS.

Audio Processing & Transmission

- Captures and encodes real-time audio streams using codecs such as **Opus or G.711**.
- Handles jitter buffering, packet loss recovery, and RTP encapsulation.
- Works closely with the signaling system to align audio start/stop with call events.
- Tests continuity and clarity under varied network scenarios to maintain high call quality.

System Output & Device Integration

- Focuses on audio playback and synchronization with delivered network streams.
- Ensures compatibility across devices (USB headsets, speakers, Bluetooth, analog outputs).
- Manages latency, consistency, and fallback behavior in case of device or stream errors.
- Provides the final rendering of decoded audio to end users with minimal distortion.

Networking, Security & Infrastructure

- Provides the backbone for reliable transport across SIP and MQTT.
- Configures and manages the **MQTT broker (Mosquitto/EMQX)** and SIP server.
- Implements **NAT traversal techniques (STUN, TURN, ICE)** for connectivity.
- Enforces security: TLS, AES encryption, access control, and client authentication.
- Simulates attacks, failure scenarios, and load testing to ensure resilience and scalability.

AI & Insights Layer

- Converts voice to text using **Whisper/Vosk**.
- Applies **NLP techniques** for summarization, meeting notes, and task extraction.
- Provides a **role-aware chatbot interface** for retrieving insights securely.
- Integrated with the signaling layer to start/stop transcription based on call state.

Management & Monitoring Layer

- Provides dashboards, logs, and alerts for performance and reliability.
- Supports system-wide testing across signaling, audio, and networking.
- Includes resilience evaluation under real-world conditions (packet loss, congestion).
- Ensure documentation, reporting, and demo readiness.

2. IMPACT

Why do you consider this project? What is its impact on community/market/end user/…?

The proposed **AI-powered Embedded PBX Communication System** will have significant impact across multiple levels:

Community Impact

Accessibility & Inclusion

- Real-time speech-to-text transcription will make communication more inclusive for individuals with hearing impairments.
- Automated meeting notes can assist community groups, NGOs, and educational institutions in documenting discussions effectively.

Digital Literacy & Skills Development

- By developing and deploying this solution locally, the project provides an example of how AI, IoT, and telecom can be integrated — inspiring students, researchers, and startups.
- It strengthens local technical expertise in cutting-edge technologies like SIP, MQTT, AI-NLP, and secure VoIP, contributing to the knowledge economy.

Support for Small Businesses & Startups

- Many small businesses and call centers cannot afford expensive enterprise communication tools.
- Our solution, based on open-source and modular components, provides a cost-effective, secure alternative that helps small organizations grow and compete.

Contribution to Egypt's Digital Transformation Goals

- The system aligns with national strategies to build a digitally capable workforce and adopt secure, AI-enabled communication infrastructure.
- By being designed and developed by Egyptian engineers, it highlights local innovation and increases confidence in homegrown technology solutions.

Potential Societal Applications

- Education: Secure virtual classrooms with automatic note-taking and role-based student/teacher access.
- Healthcare: Reliable, encrypted doctor-patient teleconsultations with AI transcription for medical records.
- Public Sector: Transparent and efficient communication in government and community services

Market Impact

- The global Unified Communications and Collaboration (UC&C) market is growing rapidly, projected to surpass USD 150 billion by 2030, fueled by AI and cloud adoption.
- In Egypt and the wider MENA region, the demand for secure, cost-effective, and intelligent communication **systems** is increasing, especially among call centers, enterprises, and public institutions.
- Current market leaders (Zoom, Microsoft Teams, Cisco Webex) are costly, not IoT-native, and do not provide role-aware AI insights, leaving a clear market gap.
- **Digital Transformation:** Provides a modern, secure, and intelligent alternative to legacy telephony infrastructure.
- **AI-Driven Innovation:** Moves beyond voice communication by embedding AI for contextual understanding, summaries, and knowledge retrieval.

- **IoT Readiness:** Enables organizations to integrate IoT devices (smartwatches, sensors, laptops) seamlessly into their communication networks.
- **Security Assurance:** Protects sensitive business conversations with end-to-end encryption, role-based access control, and intrusion detection.

End-User Impact

The AI-powered Embedded PBX Communication System is designed with a user-first approach, ensuring that employees, managers, and organizations experience tangible benefits in their day-to-day operations.

1. For Employees

- Simplified Communication: One platform for calls, video, chat, and AI insights reduces the need to switch between multiple apps.
- Productivity Boost: Automated transcription and meeting notes allow employees to focus on discussions rather than manual documentation.
- Accessibility: Real-time speech-to-text supports employees with hearing impairments, making communication more inclusive.

2. For Managers & Decision-Makers

- Actionable Insights: AI-generated summaries and task extractions provide managers with a clear view of meeting outcomes.
- Role-Based Access: Managers and admins can securely access more detailed insights, while employees see only relevant information.
- Faster Decisions: Quick retrieval of past conversations and insights through the chatbot improves decision-making speed and accuracy.

3. For Organizations

- Cost Efficiency: By unifying PBX, VoIP, AI transcription, and IoT integration into a single system, organizations save on licensing and infrastructure costs.
- Scalability: The system grows with the organization, supporting additional users, devices, and AI features without major redesign.
- Security & Compliance: Built-in encryption, RBAC, and audit logs ensure compliance with industry standards (e.g., GDPR, ISO).

4. For SMEs & Startups

- Affordable Access to Enterprise Features: Smaller businesses gain access to AI-powered communication tools usually only available to large corporations.
- Competitive Advantage: Startups and SMEs can appear more professional and efficient in client communications.

Additionally, the system promotes inclusivity, enabling those with hearing impairments or language barriers to access transcripts and use the chatbot in a text-based format.

3. NOVELTY AND FEATURES

The **AI-powered Embedded PBX Communication System** is innovative in both its **architecture** and its **functional features**, offering a level of integration not present in existing communication platforms.

A. Novelty

1. **Hybrid Signaling Model (SIP + MQTT):**
   - Unlike traditional PBX systems that rely solely on SIP, our solution combines **SIP for enterprise-grade telephony** with **MQTT for lightweight IoT integration**.
   - This enables smart devices such as smartphones, laptops, and wearables to seamlessly join the communication system.
2. **AI-Driven Insights Integrated into PBX:**
   - Existing PBX/VoIP platforms (Zoom, Cisco Webex, Microsoft Teams) focus on communication but provide limited or costly AI features.
   - Our system embeds **real-time transcription, summarization, and a role-based chatbot** natively, making it a **communication + knowledge management hub**.
3. **Security-First Architecture:**
   - Unlike many platforms where security is an afterthought, our design enforces **TLS 1.3, SRTP, RBAC, intrusion detection, and encrypted storage** across all layers.
   - Adds resilience against **AI-specific attacks** (e.g., adversarial audio, prompt injection) — a rarely addressed concern in current solutions.
4. **Localized, Affordable Innovation:**
   - By building on **open-source frameworks** (Asterisk, Mosquitto, Whisper), the system reduces licensing costs and offers **enterprise-level features at a fraction of the cost**, making it especially impactful for **SMEs and emerging markets**.

B. Key Features

1. **Unified Communication:** Voice, video, and conferencing integrated with AI insights in a single system.
2. **AI Transcription & Summarization:** Automatic generation of transcripts, meeting notes, and action points.
3. **Role-Based Chatbot:** Secure retrieval of information based on user roles (employee, manager, admin).
4. **Cross-Device Integration:** Seamless connectivity across smartphones, laptops, and IoT devices such as smartwatches.
5. **End-to-End Security:** Encryption for signaling (TLS), media (SRTP), and stored data (AES-256).
6. **Scalability & Flexibility:** Modular architecture that supports future AI models, federated learning, and integration with external systems.
7. **Management & Monitoring:** Real-time dashboards, performance logs, and security alerts for administrators.

iii) Related Products:

Product / Service	What It Offers	Similarities to Your Project	What It Doesn't Do / Gaps
Voicegain (Speech-to-Text & Telephony Bot APIs) voicegain.ai	Offers streaming/batch transcription, integration with telephony (e.g. in SIP), sentiment / analytics.	— Real-time transcription, integration with voice; useful API approach.	Doesn't provide full PBX + IoT signaling + hybrid model; likely doesn't have MQTT + device integration + role-based chatbot as core.
Elevate UC elevatetechnology.com	Cloud phone system that integrates voice, video, chat, SMS, includes AI features (voicemail transcription, meeting summaries).	Unified communication + AI summaries.	Probably no MQTT / IoT device integration; might depend mostly on cloud; less likely that signaling flexibility (custom topic hierarchies, hybrid gateway) is open.
Epygi AI-Powered Solutions Epygi	Provides transcription of voicemails, call recordings, some AI features.	Works with call recording; adds intelligence to voice systems.	Again, doesn't cover full live, hybrid signaling, or real-time IoT device signaling; might lack role-aware chatbot etc.
TruSIP AI Powered SIP Trunks truSIP	SIP trunks with automatic call transcription, analytics, summaries, sentiment, etc.	Good match on the telephony + AI side.	Doesn't integrate MQTT or IoT bots; less flexibility in device types; might not provide full media + signaling hybrid flexibility.
Nautilus Call Transcription & Sentiment Analysis nautilus-network.com	Transcribes calls and does sentiment analysis; integrates into reporting tools.	Strong in AI analytics, summarization.	Probably reactive (after call), not necessarily embedded real-time or multi-device + IoT + signaling control.

4. DELIVERABLES
What is the project final outcomes (HW device, SW package, simulation ...)?

The Final deliverables of this project will be:

DELIVERABLES

By the end of the project, the team will provide the following deliverables:

1. Functional Prototype

- A working AI-powered Embedded PBX system that integrates:
  - Hybrid signaling (SIP + MQTT) for call setup and management.
  - Media layer for secure voice, video, and conferencing.
  - AI module for real-time transcription, summarization, and chatbot-based retrieval.
  - Security module implementing TLS, SRTP, RBAC, and intrusion detection.

2. AI & Chatbot Demonstration

- Live demonstration of:
  - Real-time transcription of voice calls.
  - Automated meeting notes generation.
  - Role-based chatbot that retrieves insights differently for employees, managers, and admins.

3. Documentation

- Technical Report: Detailed description of system design, architecture, protocols (SIP, MQTT), and AI pipeline.
- Security Report: Documentation of applied security measures, testing results, and compliance considerations.
- User Manual: Guide for employees and administrators to operate the system.
- Installation & Deployment Guide: Steps for replicating the system on servers or cloud environments.

4. Testing & Evaluation Results

- Results from stress-testing signaling protocols (SIPp, MQTT stresser).
- Media quality reports (latency, jitter, packet loss analysis).
- Security penetration testing results showing resilience to attacks.
- AI performance evaluation (accuracy of transcription, quality of summaries).

5. Management & Monitoring Dashboard

- A visual dashboard for administrators, showing:
  - Call/session states.
  - AI insights usage.
  - System health and logs.
  - Security alert

5. BUSINESS PLAN AND MARKET ANALYSIS

A) Business Plan

1. Value Proposition

The proposed system is an AI-powered Embedded PBX communication platform that unifies:

- VoIP (SIP) reliability,
- IoT (MQTT) scalability,
- AI-driven transcription and insights, and
- Enterprise-grade security.

It eliminates the need for organizations to subscribe to multiple fragmented tools (PBX, transcription software, note-taking apps, chatbot assistants) by delivering an all-in-one secure platform.

2. Business Model

- Licensing Model: Offer the solution as an on-premise PBX + AI package for enterprises.
- Subscription Model (SaaS/UCaaS): Cloud-hosted service for SMEs, billed per user/month.
- Customization & Integration Services: Tailored deployments for call centers, banks, healthcare, or public-sector organizations.
- Support & Maintenance: Annual support contracts for security updates, feature enhancements, and monitoring.

3. Revenue Streams

- **Software licenses/subscriptions.**
- Professional services (integration, customization).
- Premium AI modules (advanced analytics, predictive insights, sentiment tracking).
- Enterprise support packages.

4. Go-to-Market Strategy

- Initial Target: Call centers and SMEs in Egypt, where cost and efficiency are critical.
- Expansion: Enterprises in MENA and Africa seeking secure AI-enhanced telephony.
- Partnerships: Telecom providers (for SIP integration) and local cloud providers (for SaaS hosting).
- Pilot Programs: Deploy pilot systems in universities, hospitals, or call centers to showcase ROI.

B) Market Analysis

1. Market Size & Opportunity

- The global UCaaS (Unified Communications as a Service) market is projected to reach USD 150+ billion by 2030, driven by AI and remote work.
- The Egyptian call center industry is one of the largest in MENA, employing tens of thousands and requiring scalable, cost-efficient telephony.
- Growing IoT adoption in enterprises creates demand for lightweight integration with wearables, laptops, and mobile devices.

2. Competitor Landscape

- Global Competitors: Zoom, Microsoft Teams, Cisco Webex → focus on communication, but AI is limited or premium, and IoT integration is absent.
- Niche Competitors: Voicegain, Epygi, TruSIP → provide telephony with AI features, but lack a hybrid SIP + MQTT architecture and role-aware AI insights.
- Differentiation: Our system provides hybrid signaling, embedded AI, IoT-native design, and security-first architecture, making it unique.

3. Target Market Segments

- Call Centers & BPOs: Need scalable, AI-enhanced telephony at lower cost.
- SMEs: Require affordable all-in-one solutions without managing multiple tools.
- Enterprises & Banks: Need compliance-ready, secure systems with role-based insights.
- Public Sector & Education: Demand secure conferencing, automated notes, and knowledge management.

4. Growth & Scalability

- Short-term: Prototype → pilot deployments in SMEs/call centers.
- Mid-term: Scale to SaaS model for broader adoption.
- Long-term: Regional expansion across MENA, offering exportable Egyptian technology.

6. GANTT CHART

List your project's phases and tasks down the side of your chart.

Tasks/Activities	Month # 1	Month # 2	Month # 3	Month # 4	Month # 5	Month # 6	Month # 7	Month # 8	Month # 9	Month # 10
Requirement Analysis & Research	✓	✓								
System Architecture Design	✓	✓								
MQTT-Based Audio Communication Setup		✓	✓	✓						
AI Module (STT + NLP) Development			✓	✓	✓					
Role-Based Chatbot Assistant				✓	✓	✓				
UI Development (Web/Mobile Dashboards)					✓	✓	✓			
IoT Device Sync & Smart Integration						✓	✓	✓		
Security & Data Protection Layer						✓	✓			
Integration Testing & Debugging								✓	✓	✓
Documentation & Reporting								✓	✓	✓
Final Demo & Presentation Preparation									✓	✓

7. List of needed HW/SW Components					
Item	**Type (Hardware/ Software/ Other)**	**Specifications (brief description)**	**Unit Cost**	**No. of Items**	**Total Cost of Items**
1	ESP32 / Raspberry Pi	Embedded device for audio processing and MQTT communication	4000	2	8000
2	MQTT Broker (Mosquitto)	**Lightweight message broker for communication**	free		free
3	ESP32 Module	IoT node for device communication and sensor integration	300	4	1200
4	Sensors	That need in our target output			
5	SQLite / MongoDB	**Database for storing transcripts and summaries**	free		free
6	Hosting (optional VPS)	For testing deployment and API access	500/month	3 months	1,500
7					
8					

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Cite to the references those related to your idea

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